

各控制信号的逻辑表达式为：

$$MA = W1T2 + W1T3 (MOV1 + MOV2 + MOV3)$$

$$RA = W2[T1(ADD + SUB + MUL + DIV + AND + OR + NOR + XOR + ADD_SELF + SUB_SELF + LSL + LSR) + T4MOV3]$$

$$PB = W1(T1 + T3) + W2(T1 + T2)(MOV1 + MOV2 + MOV3)$$

$$RB = W2T1(ADD + SUB + MUL + DIV + AND + OR + NOR + XOR)$$

$$CPR0 = W2[T3MOV1 + T1(ADD + SUB + MUL + DIV + AND + OR + NOR + XOR + ADD_SELF + SUB_SELF + LSL + LSR)]$$

$$CPR1 = W2T3MOV2$$

$$CPPC = W1T3 + W2T2(MOV1 + MOV2 + MOV3)$$

$$CPIR = W1T2$$

$$CPMAR = W1T1 + W2[T1(MOV1 + MOV2 + MOV3) + T3MOV3]$$

$$RD = W1T2 + W2T3(MOV1 + MOV2 + MOV3)$$

$$WR = W2T4MOV3$$

$$G = W2T1HALT$$

$$M = W1\overline{T3} + W2\overline{T2}(MOV1 + MOV2 + MOV3) + W2\overline{T1}(ADD + SUB + ADD_SELF + SUB_SELF) + W2T1(AND + OR + NOR + XOR)$$

$$S3 = W1 + W2(MOV1 + MOV2 + MOV3 + ADD + AND + OR + SUB_SELF) + W2\overline{T1}(SUB + NOR + XOR + ADD_SELF)$$

$$S2 = W1\overline{T1}\overline{T3} + W2\overline{T1}\overline{T2}(MOV1 + MOV2 + MOV3) + W2\overline{T1}(ADD + AND + NOR + ADD_SELF) + W2T1(SUB + OR + XOR + SUB_SELF)$$

$$S1 = W1\overline{T3} + W2\overline{T2}(MOV1 + MOV2 + MOV3) + W2\overline{T1}(ADD + NOR + ADD_SELF) + W2T1(SUB + AND + OR + XOR + SUB_SELF)$$

$$S0 = W1\overline{T1} + W2\overline{T1}(MOV1 + MOV2 + MOV3 + SUB + OR + XOR + ADD_SELF) + W2T1(ADD + AND + NOR + SUB_SELF)$$

$$CN = W1\overline{T3} + W2\overline{T2}(MOV1 + MOV2 + MOV3) + W2T1(ADD + SUB_SELF)$$

+W2 $\overline{T2}$ (SUB+ADD_SELF)

S8 = W2T1DIV

S7 = W2T1(LSL + LSR)

S6 = W2T1(LSL + LSR)

S5 = W2T1LSL + W2 $\overline{T1}$ LSR

S4 = W2T1MUL